Prediction Markets vs Rates Markets: A Statistical Arbitrage Framework for Fed Expectations

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Abstract

We develop a cross-market framework to study disagreement in FOMC policy expectations across prediction markets (Polymarket, Kalshi) and institutional rates markets (Fed Funds futures, SOFR futures, SOFR OIS). We evaluate two strategies: a pure arbitrage spread trading prediction-market versus rates-implied expected moves, and a statistical arbitrage approach using PCA residuals with Ornstein–Uhlenbeck mean-reversion. The pure spread strategy yields positive but modest results across 83 trades ($2,617.55 net P&L, per-trade Sharpe of 0.47), confirming the presence of a converging spread but also significant transaction costs. The statistical arbitrage framework identifies a robust mean-reversion signal, but one that cannot be traded profitably in the presence of high transaction costs. Overall, cross-venue disagreement in FOMC expectations may contain tradable components, though scalability remains constrained by prediction-market liquidity and transaction costs.

We have switched the recommended order of our Method and Methodology, as we believe this makes more sense for our report.

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1 Introduction

Prediction markets (Polymarket, Kalshi) and CME Fed Funds futures both price FOMC policy decisions, but attract different participants, operate under different collateral regimes, and have different microstructure conventions. If both are efficient, they should agree on the expected policy outcome once translated into common units. In practice they do not always agree. We ask whether the disagreement is exploitable, comparing event-contract probabilities from Polymarket and Kalshi [1, 2] with rates-implied expectations from Fed Funds futures, SOFR futures, and SOFR OIS [3, 4, 5].

The informational value of prediction markets for monetary policy has recently been validated by Diercks, Katz and Wright [6], who show that Kalshi-implied distributions are well-calibrated, rapidly updating, and competitive with professional forecasters for FOMC rate decisions, in several cases outperforming Fed Funds futures outright. Their findings establish that prediction markets are a credible source of policy expectations, and motivate the natural follow-up question we address here: if prediction markets and institutional derivatives both embed reliable information, why do they sometimes disagree, and can that disagreement be traded?

Our contribution is an end-to-end, reproducible pipeline that converts both instrument families into a common state variable: the expected basis-point rate change at each meeting. It trades the cross-venue spread in two ways. First, a pure arbitrage strategy that enters when the spread exceeds explicit transaction costs and holds to expiry, locking in the difference. Second, a statistical arbitrage strategy that looks to extract common factor variation via Principal Component Analysis (PCA) and trades mean reversion in the residual spread using Ornstein-Uhlenbeck rules, following the framework of Avellaneda and Lee [7].

2 Methodology

2.1 Overview

Our methodology builds a meeting-date panel that aligns prediction-market probabilities and rates-market implied expectations in common units of basis-point policy changes, then evaluates two strategy classes under explicit transaction costs. The design is carefully implemented to avoid look-ahead bias: signals at date t use only information available by date t. OU parameters are lagged one period before use, PCA is fitted on an expanding window, and all imputation uses strictly backward-looking information.

2.2 Data

The sample spans May 2023 to January 2026, covering 39 FOMC meetings with Kalshi data and 30 with Polymarket data. In this section we will present the different data sources used to create the arbitrage strategies in this report.

2.2.1 Polymarket and Kalshi FOMC Contracts

Polymarket and Kalshi list binary event contracts on discrete FOMC outcomes. Each contract settles at \$1 if the outcome occurs and \$0 otherwise. Prices are interpreted as risk-neutral probability proxies in $[0, 1]$. Raw files contain 9,071 rows; the merged panel covers 2,318/2,484 rows (93%) across 30 meetings.

Kalshi lists analogous binary FOMC contracts. Raw files contain 15,759 rows; the merged panel covers 1,582/2,484 rows (64%) across 24 meetings, with sparse coverage before 2024. The dataset includes `close_bid` and `close_ask` fields used to estimate empirical bid-ask spreads.

2.2.2 CME 30-Day Fed Funds Futures

CME Fed Fund futures settle to $100 - \bar{R}_M^{EFFR}$, the arithmetic average of daily EFFR over calendar month M. Daily settlement prices and volumes are sourced from CME.

2.2.3 CME One-Month SOFR Futures (SR1)

SR1 futures settle to $100 - \bar{R}_M^{SOFR}$, the arithmetic average of daily SOFR over calendar month M, and are structurally analogous to Fed Fund Futures. The SOFR-EFFR basis is treated as approximately constant over short horizons. Daily settlement prices and volumes are sourced from CME with near-dated coverage from 2022 onward.

2.2.4 SOFR Overnight Index Swaps (OIS)

SOFR OIS quotes are sourced at standard tenors (1W, 2W, 1M, 2M, 3M, 6M). Data coverage begins in early 2023. Tenors beyond 6 months are excluded as they span too many meetings to provide reliable local signal.

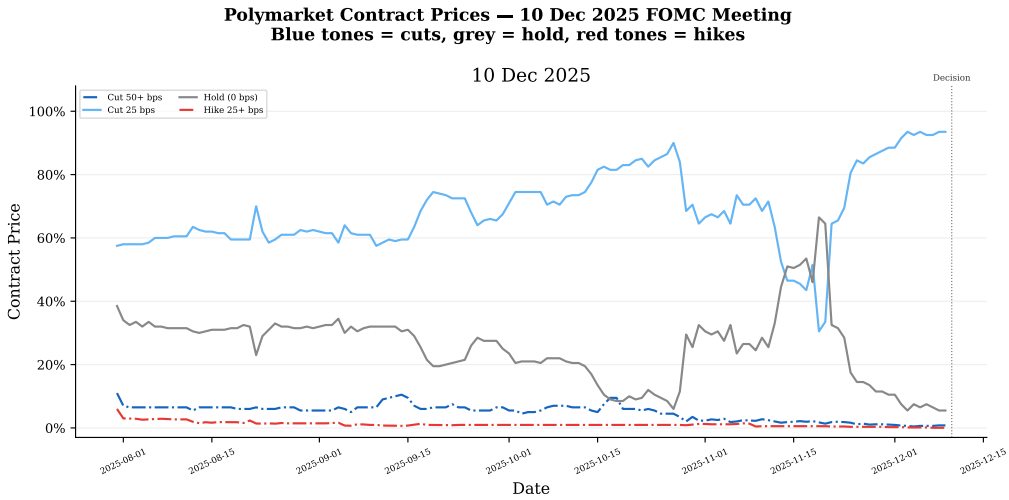


Figure 1: Polymarket contract probabilities (which are also a proxy of their price) converging to the true value, for the Fed meeting decision on 10 December 2025. The decison was to cut by 25bps.

2.3 Data Engineering and Alignment

The merged panel standardizes timestamps for EFFR, SR1, Kalshi and Polymarket to a common daily snapshot, at 2pm CST/CDT. EFFR and SR1 settlements are printed daily by CME at 2pm CT, made publicly available within minutes. Polymarket and Kalshi data is streamed live and accessible via API at minute-by-minute intervals. Polymarket and Kalshi data is obtained directly and data aligned to 2pm CT is retained. OIS data is available at of close, 5pm CT. As a result, the merged panel for EFFR, SR1, Polymarket and Kalshi represents aligned and immediately tradeable data, noting that all four relevant markets are open at 2pm CT. Missing values are handled with strictly backward-looking imputation: forward carry within each meeting's time series, then backward fill using the historical mean up to but not including the current date. Full implementation details are in Appendix B.2.

3 Model I: Pure Arbitrage

3.1 Betting on the Rate Decision

For each FOMC meeting m and observation date t, let H_m denote the realized policy-rate change (in basis points) associated with the decision. The latent object of interest is the conditional expectation $\mathbb{E}_t[H_m]$. The modeling problem is to construct multiple estimators of this quantity from different markets, quantify their disagreement, and determine whether trade opportunities exist.

3.2 Prediction-market: Implied Rate Change

Prediction contracts are binary claims that settle at \$1 if a specific outcome occurs and \$0 otherwise. Let $p_{i,t}$ denote the contract price (interpreted as a probability proxy) for bucket i observed on date t for some fed decision, and let h_i denote the bucket's assigned policy move in basis points. The expected move estimator is

$$\widehat{H}_t^{PM} = \frac{\sum_i h_i \, p_{i,t}}{\sum_i p_{i,t}}.$$

Normalization by total quoted mass is essential because bucket menus are occasionally incomplete, so raw bucket prices may not sum to one. Under the assumption that unquoted outcomes have zero probability, this normalization rescales the available probabilities to sum to one while preserving relative probability ratios.

The assignment of a scalar basis-point value to each bucket follows a canonical mapping. For standard-size buckets (`C25`, `H0`, `H25`, `H50`), the assignment is unambiguous. For open-ended tail buckets (`C50+`, `H50+`, `C75+`, etc.), we assign the boundary value. In practice the probability is so small that this has a negligible effect.

3.3 CME 30-Day Fed Funds Futures: Implied Rate Change

CME 30-Day Fed Funds Futures (`ZQ`) settle at 100 minus the arithmetic average of the daily EFFR over the delivery month. Let r_m denote the futures-implied average EFFR for calendar month m:

$$r_m = 100 - S_m \tag{1}$$

where S_m is the settlement price of the contract expiring at the end of month m.

Let month m contain a decision on day d out of D total days. Let r^- denote the prevailing overnight rate before the decision and $r^+ = r^- + H$ the rate after, where H is the announced change. The monthly average is a weighted average of the pre- and post-decision rates:

$$r_m = \frac{d}{D} r^- + \frac{D-d}{D} r^+ \tag{2}$$

Substituting $r^+ = r^- + H$ and letting $\lambda = d/D$:

$$r_m = \lambda r^- + (1-\lambda)(r^- + H) \tag{3}$$

$$= r^- + (1-\lambda)H \tag{4}$$

Taking expectations:

$$E[r_m] = E[r^-] + (1-\lambda)\, E[H] \tag{5}$$

If month $m-1$ contains no decision, the overnight rate is constant throughout that month at r^-, so:

$$E[r_{m-1}] = E[r^-] \tag{6}$$

Substituting into (5) and solving:

$$\widehat{H}^{\text{CME}} = \frac{r_m - r_{m-1}}{1-\lambda} \tag{7}$$

where r_{m-1} and r_m are read directly from the settlement prices of the two corresponding futures contracts.

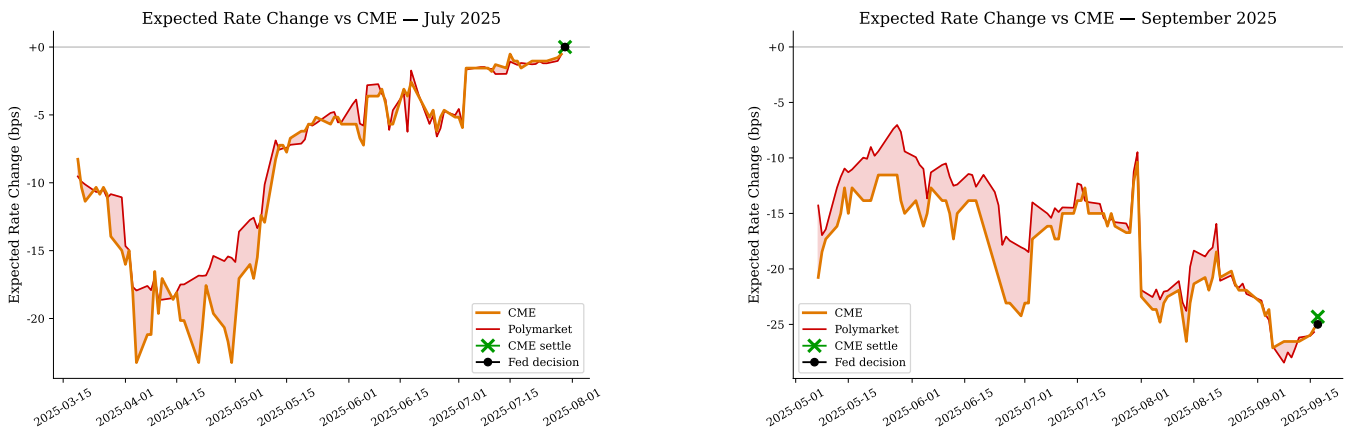


Figure 2: Plots of the expected rate change using both instruments we have constructed, with their spread shaded in red, as well as where they expire.

3.4 Portfolio Construction

3.4.1 DV01 of a Fed Funds Futures Contract

The actual payout for a fed futures contracts is scaled to mirror the interest earned over one month with interest rate $\bar{R}$. The dollar value of a 1 basis point move is

$$\text{DV01} = N \times \frac{d}{360} \times \frac{1}{10{,}000} = \frac{N\,d}{3{,}600{,}000} \tag{8}$$

where $N = \$5{,}000{,}000$ is the contract notional and d is the number of calendar days in the delivery month (actual/360 day-count convention). For a 30-day month:

$$\text{DV01} = \frac{5{,}000{,}000 \times 30}{3{,}600{,}000} = \$41.67 \tag{9}$$

We cannot buy a fraction of a future. Therefore to make the instrument defined in equation 7, we long/short two futures which correspond to months m and $m-1$. This represents the value $(1-\lambda)\widehat{H}^{\text{CME}}$, giving us the final capital allocation:

$$C_{\text{CME}} = N^{\text{CME}} \cdot \text{DV01} \cdot (r_m - r_{m-1}) \tag{10}$$

Where N^{CME} is some integer number of contracts that we buy and DV01 was used to convert from bps to dollars. In our backtest we will buy CME contracts one at a time so we set $N^{\text{CME}} = 1$.

3.4.2 Polymarket Replicating Portfolio

At expiry, the Fed announces outcome k, where $k \in \{h_i\}$ is the realized policy move in basis points. The combined P&L of the two legs under the arbitrage assumption is:

$$\Pi_k = \underbrace{N^{\text{CME}} \times h_k \times \text{DV01}}_{\text{CME leg}} - \underbrace{N_k \times \$1}_{\text{PM leg}} = 0 \tag{11}$$

This is satisfied exactly when:

$$N_k = h_k \times \text{DV01} \tag{12}$$

Where we have set $N^{\text{CME}} = 1$ as previously mentioned. That is, for each outcome bucket k, hold $h_k \times \text{DV01}$ prediction market contracts.

As previously explained one CME portfolio pays $\text{DV01} \times (1-\lambda)\, H_0$. To match this on Polymarket, we replicate the payoff $(1-\lambda)\, H_0$, which requires $\text{DV01} \times (1-\lambda) \times |h_i|$ contracts on each outcome i, buying when $h_i > 0$ and selling when $h_i < 0$. The total capital allocated is:

$$C_{\text{PM}} = \text{DV01} \cdot (1-\lambda) \cdot \left(\sum_{\{i:\, h_i > 0\}} h_i\, p_i + \sum_{\{i:\, h_i < 0\}} |h_i|\, (1-p_i) \right) \$ \tag{13}$$

The first sum is the cost of buying the hike-side contracts, the second is the cost of selling (buying the complement of) the cut-side contracts. We provide a detailed discussion of the $(1-\lambda)$ factor in the Appendix. Its effect can be essentially be ignored in this report, but it would have to be monitored in a full scale implementation of this strategy.

3.5 Trading Protocol

We define the spread as,

$$S_t = \widehat{H}_t^{PM} - \widehat{H}_t^{CME}, \tag{14}$$

dropping the month index and noting that every meeting will correspond to a spread. With the portfolios we have constructed, $\widehat{H}_t^{PM}$ and $\widehat{H}_t^{CME}$ will in theory converge to the true value at expiry of both contracts: the actual rate cut after FOMC meeting. In practice, this may not always be true (see section 3.6.1).

Entering the trade: We enter a trade whenever the spread exceeds the transaction costs (TC)[1]. Based upon the assumption that prices converge, this will lock in a guaranteed profit of $|S_t - \text{TC}_t|$. Thinking of the spread as a new trading instrument itself, we short the spread if it is overpriced and long it if underpriced.

In our backtest, we found that Kalshi's transaction costs were always too high, notably because of their fee structure, which polymarket does not possess. From here onwards, we will only use polymarket in our backtesting for this strategy.

After the trade has been entered, we wait until expiry and are paid the spread. If any other signal appears, we enter again.

3.6 Limitations

3.6.1 Expiry at a Non-Zero Spread

The offset between the target rate floor and the daily effective rate is usually constant. There were two consecutive months in recent history where the assumption broke, in which the offset climbed at a roughly linear rate. The total gain was 4 basis points in each month, increasing the average of each month by 2 basis points. Depending on whether we are long or short, this can either help or hurt the strategy.

3.6.2 Tail Clipping

The prediction markets do not offer a full distribution. Instead, due to the rarity of extreme decisions, tails are grouped together as in H50+ and C50+. We will treat H50+ as H50, as that is where most of the actual probability is concentrated. Historically, moves of 75+ are reserved for extreme macroeconomic conditions, and under such circumstances prediction markets will almost certainly update the structure to allow for specific bets on H50 and H75+, or H50, H75, H100+. Our approach flexibly integrates all markets offered for each decision.

[1] These are explained in full detail in section 7. Note that since we do not have to make a round trip, we pay less in transaction costs than in a normal statistical arbitrage set-up.

3.6.3 Market impact on Prediction Markets

The binding constraint is not the capital but the liquidity of the order book. Let D_i denote the depth (in contracts) available at or near the quoted price p_i for outcome i. The replicating portfolio requires DV01 $\times$ $|h_i|$ contracts on outcome i. If this exceeds D_i, the marginal contracts fill at worse prices and the effective E_{PM} moves toward E_F, shrinking the spread. In practice, the thinnest outcome book is the bottleneck. In this report we will omit any detailed analysis of liquidity, based on the assumption that we are buying a reasonable number of contracts such that they are available, or at least replenish quickly.

3.6.4 Buying a Single Future

In the backtest we only ever buy one CME contract at a time. CME impact cost scales as $N_{\text{contracts}}$; Polymarket TC scales linearly in $N_{\text{contracts}}$. At realistic trade sizes (e.g. 10–50 contracts) the Polymarket transaction costs would be the binding constraint.

4 Results I: Pure Arbitrage

We present the analytics collected from running the pure arbitrage strategy over all decision months which had good quality data. We have defined the Sharpe ratio per trade (not annualized) as

$$\text{Sharpe}_{\text{per trade}} = \frac{\bar{r}}{\hat{s}_r}$$

where $\bar{r} = \frac{1}{N}\sum_{i=1}^{N} r_i$ is the mean net P&L per trade, $\hat{s}_r$ is the sample standard deviation, and r_i is the net P&L of trade i in USD.

We make an important note that performance analytics in this section are less meaningful than in the statistical arbitrage setting. Since this is a pure arbitrage strategy, one could in principle scale positions arbitrarily to maximise profits — subject to practical constraints such as market impact, liquidity, and transaction costs. In this study we restrict ourselves to the minimum tradeable size on entry, which removes liquidity concerns entirely. Scaling up would be straightforward in principle but would require careful analysis of the constraints above. The results here should therefore be interpreted as a proof of concept demonstrating that the arbitrage opportunity exists, rather than as a realistic estimate of achievable returns at scale.

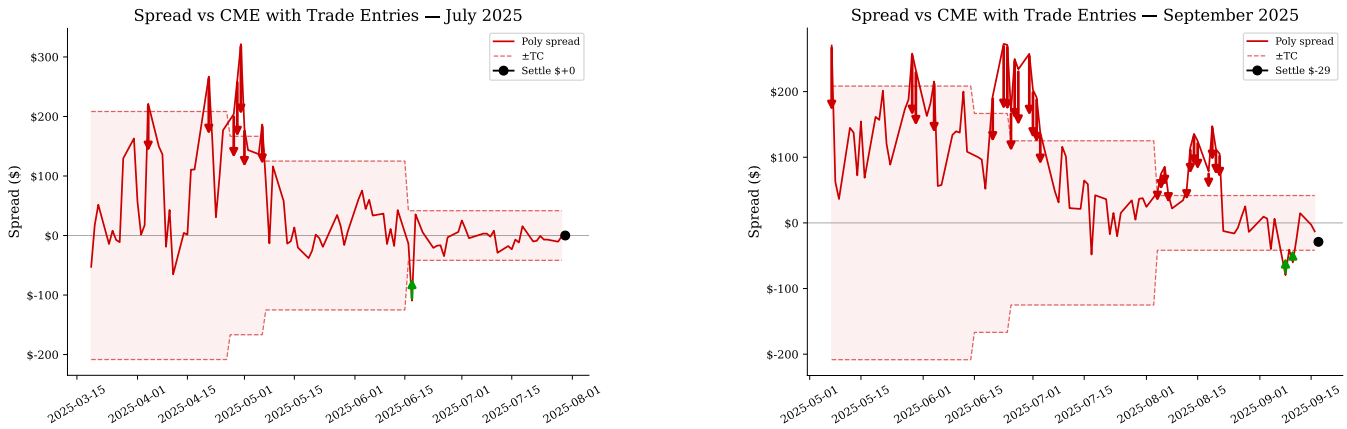


Figure 3: The spread over time (in USD) is plotted with trade entry points marked by color (red for long, green for short). Notably, the September 2025 spread does not settle to zero, illustrating one of the limitations discussed in section 3.6.1. Full plots for all available decision months are provided in the Appendix.

5 Model II: Statistical Arbitrage Strategy

5.1 Overview

Model II implements a factor-residual statistical arbitrage framework. It combines meeting-jump expectations estimated from rates instruments with prediction-market features, utilizes PCA to remove common components, and trades mean reversion in the residuals using Ornstein-Uhlenbeck (OU) rules, following Avellaneda and Lee (2010) [7].

Metric	Value
Total trades	83
Wins / Losses	56 / 27
Win rate	0.7%
Total gross PnL	\$+9,785.51
Total TC (PM spread)	\$7,167.95 (70% of gross)
Total net PnL	\$+2,617.55
Average gross / trade	\$117.90
Average net / trade	\$+31.54
Std dev net / trade	\$66.54
Sharpe ratio (per trade)	+0.47
Average days held	51.7

Table 1: Overall backtest summary for the pure arbitrage strategy method.

5.2 Rates-implied jump extraction

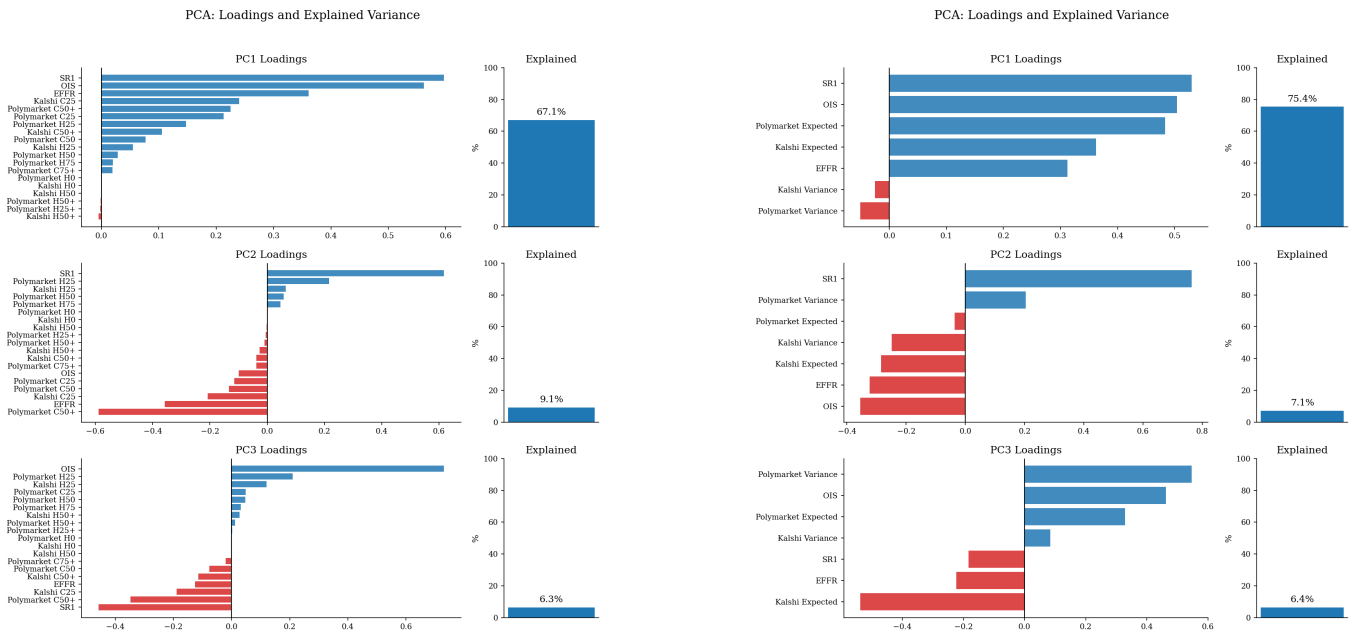
To form the panel of institutional rates, estimators based on 1-month SOFR Futures (SR1) and SOFR OIS swaps are constructed in addition to the CME Fed Funds estimator used above. Estimation of expected jumps using SOFR Futures and OIS Swaps is complicated by funding irregularities and varied tenors, and as a result we extract meeting jumps using a linear system rather than a naive two-contract decomposition. At each date t, instrument rates are represented as

$$y_t = A_t \theta_t + \varepsilon_t,$$

where A_t encodes post-meeting exposure fractions and θ_t contains baseline and meeting-jump parameters. Estimation uses ridge-stabilized least squares with a small penalty applied to jump parameters: ($\lambda = 10^{-6}$). The resulting jump estimates are directly replicable via pseudoinverse portfolio weights used downstream in execution mapping. Full SR1/OIS construction details, tenor parsing, calendar logic, and weight extraction are in Appendix D.1.

5.3 Feature panel and residual construction

The full tradable panel therefore combines the three institutional assets (EFFR, SR1, OIS) with prediction-market derived features. PCA is then fitted to the centered features, retaining $K = 3$ components. The loadings are visualized below for two panel configurations: (i) the full set of Polymarket and Kalshi assets; and (ii) composite moment estimators based on the underlying assets.



The first principal component is directly interpretable for both decompositions as the shared factor across institutional and prediction markets. This is analogous to the "market factor" in equity PCA, but here captures the common macro

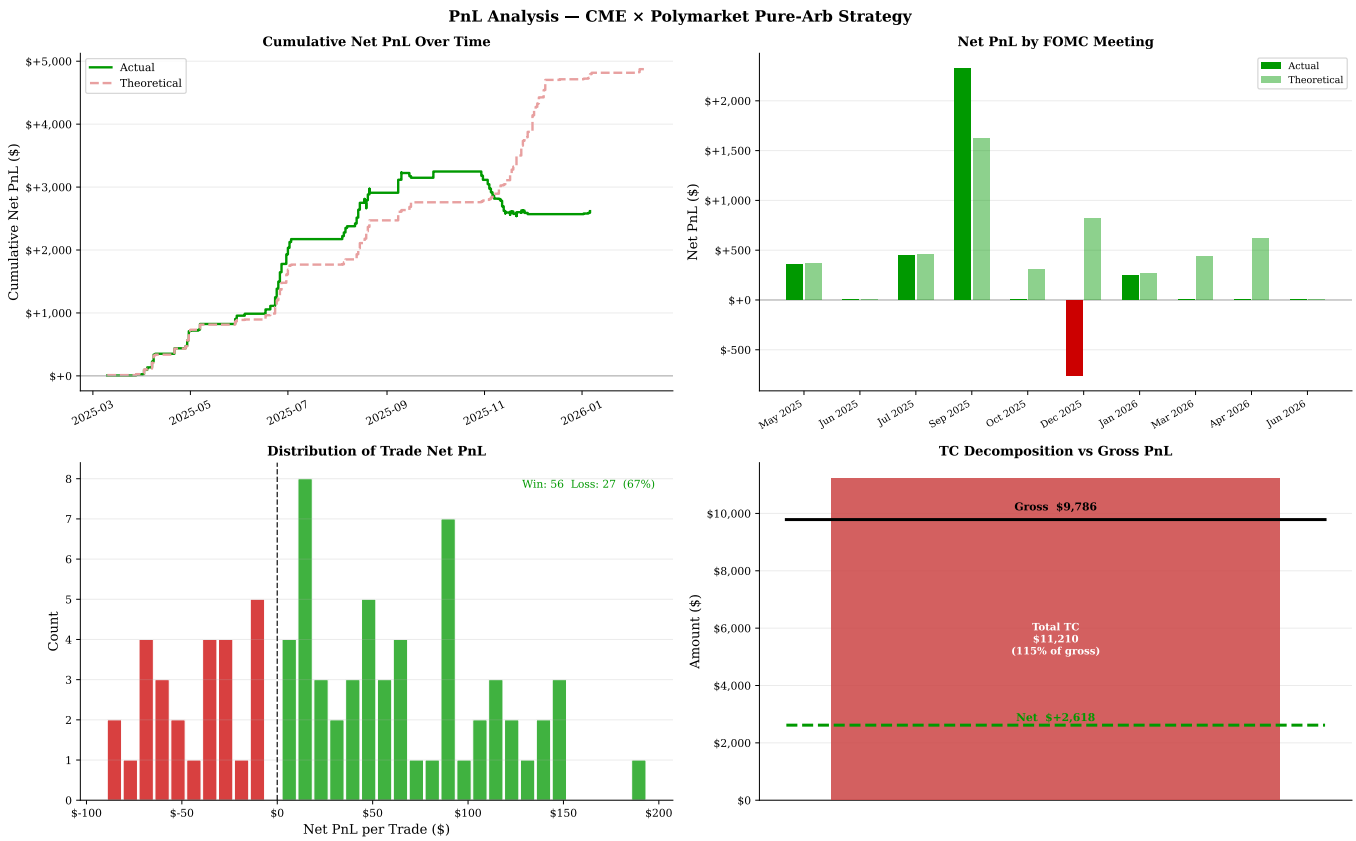


Figure 4: The table presents trade-level P&L for the pure arbitrage strategy. The theoretical P&L assumes the spread expires at exactly zero, representing the idealized case where both markets converge perfectly at settlement. In practice, realized P&L can exceed this benchmark. This occurs when the spread moves in our favour during the holding period, increasing the effective spread captured at expiry even if final settlement is not exactly zero. A notable feature of the results is the significant drag from transaction costs (bottom right), which constitute a substantial fraction of total gross revenue. This highlights why the strategy is only viable above a minimum spread threshold, as discussed in the trading protocol. The poor performance in December is attributable to the EFFR offset phenomenon described in Section 3.6.1: the offset drifted during the month, causing the spread to expire at a non-zero value and resulting in a loss.

policy expectation across venues. It explains most of the variation (about 75% in the moments panel). The second component appears to load on dispersion/tail-probability structure: in the full prediction-market decomposition, tail buckets (for example, Polymarket C50+) load with opposite sign to central buckets.

Although $K = 3$ components are fitted, trading residuals are constructed by removing only the first ("market") component. Removing only the dominant shared factor, rather than all three, preserves the potentially tradable idiosyncratic information embedded in the remaining variation, including systematic differences between how retail prediction-market participants and institutional rates traders form and revise their policy expectations. Removing additional factors would progressively whiten the residual toward noise.

The full definitions for the asset decompositions, imputation rules and PCA residual formulas are provided in Appendix D.2.

5.4 OU signal rules and tradability filters

Residuals are modeled using a rolling AR(1) approximation of an OU process. Parameters are lagged by one period to prevent look-ahead bias. Following Avellaneda and Lee (2010), baseline thresholds are:

- Entry/exit bands: $c_{\text{entry}} = 1.25$, $c_{\text{exit}} = 0.5$.
- Mean-reversion filter: OU half-life ≤ 15 days.

Positions are forced flat within 2 days of meeting to avoid possibly high volatility periods in proximity to the meetings. Complete OU formulas and filter hierarchy are in Appendix D.3.

5.5 Execution mapping and P&L accounting

Signals are converted to tradable baskets in two layers: (i) PCA residual weights across feature assets and (ii) feature-to-underlying mappings (EFFR contracts, SR1/OIS estimator weights, and prediction buckets). Daily mark-to-market P&L is computed from entry-locked prices and current closes, then aggregated across active positions and meetings. Sharpe is annualized using 365 periods for calendar-day panels and 252 for business-day-only panels [8].

Exact mapping equations, execution-lag handling, and P&L aggregation conventions are in Appendix D.5.

6 Results II: Statistical Arbitrage

The key empirical finding is that while the PCA-residual OU signal is statistically robust in gross terms, it is not economically tradable once realistic execution frictions are applied.

Table 2 shows results from running the statistical arbitrage strategy across various sensitivities.

The first four columns (Moments, All, Grouped, and Expected) correspond to different transformations of the underlying prediction market data as described in Appendix D.2. Lag 1 implements a one-day execution lag and includes OIS data, which is only available with an approximately three-hour reporting delay; for panels that exclude OIS, same-day execution is feasible as discussed in Section 2.

Restricted I tightens by filtering for OU half-life ≤ 5 days, identifying signals with stronger mean-reversion tendency. Restricted II additionally requires the entry band to $c_{\text{entry}} = 2.5$ standard deviations (from the baseline of 1.25), requiring a stronger mean-reversion signal before entering a position. Both restrictions substantially reduce the number of trades and, consequently, total transaction costs. Notably, gross signal quality improves under the restricted specifications (gross Sharpe and win rate both rise), confirming that the mean-reversion signal is real and becomes cleaner when high-conviction conditions are required.

Despite the tighter filters, transaction costs still outweigh total gross profit by almost an order of magnitude across all variants.

Metric	**Moments**	**All**	**Grouped**	**Expected**	**Lag 1**	**Restricted I**	**Restricted II**
Gross Sharpe Ratio	0.1308	0.1361	0.9681	0.4834	-0.3239	1.1297	1.2968
Net Sharpe Ratio	-7.2768	-9.5270	-10.4375	-11.1048	-8.1511	-9.1969	-7.5300
Gross Total Profit	23.9680	19.7338	149.2430	120.6807	-49.8619	162.2672	112.4910
Net Total Profit	-1588.0788	-2151.4227	-2977.4011	-4899.4637	-1618.6138	-2060.1682	-838.1819
Total Txn Costs	1612.0468	2171.1565	3126.6441	5020.1444	1568.7519	2222.4354	950.6729
Gross Win Rate	0.5253	0.5658	0.5714	0.5813	0.5031	0.5938	0.6074
Net Win Rate	0.1020	0.0463	0.0278	0.0256	0.0981	0.0339	0.0368
Number of Trades	990	562	504	664	968	384	163
Avg Trade Length (days)	6.5768	6.1708	6.0317	6.1039	6.5331	6.2708	6.2577

Table 2: Performance comparison across strategy variants

Further investigation shows that gross Sharpe is positive for every asset class considered, including CME-linked legs and both prediction-market venues. This supports the core claim that the methodology identifies a consistent directional-reversion signal in residual space.

The cost drag is dominated by prediction-market execution (Kalshi and Polymarket spread and fee assumptions), with CME costs materially smaller in comparison.

6.1 Sensitivity analysis

We tested a range of alternative specifications around the baseline implementation: stronger entry thresholds, alternative panel constructions (including richer prediction features), and parameter variations in OU/PCA windows and holding constraints. Across these variants, gross performance moves but the qualitative conclusion is unchanged: realistic transaction costs, especially in prediction markets, eliminate net profitability.

We conclude that alpha exists at the signal level, but current market microstructure costs fully absorb it at tradable size. This motivates the use of the pure-arbitrage design in which transaction costs are built directly into opportunity definition and trade selection, rather than treated as a hurdle the statistical signal must overcome after implementation.

7 Transaction costs

Transaction costs are modeled separately by venue. **For prediction-market legs:**

- **Bid–ask spread:** Using Kalshi bid–ask data, we approximate the bid–ask spread as a function of days to decision (5). We use 4-rule approximation for both Kalshi and Polymarket spread costs: Bid–ask spread = 0.05 (days > 95), 0.04 (85 < days ≤ 95), 0.03 (45 < days ≤ 85), 0.01 (days ≤ 45).

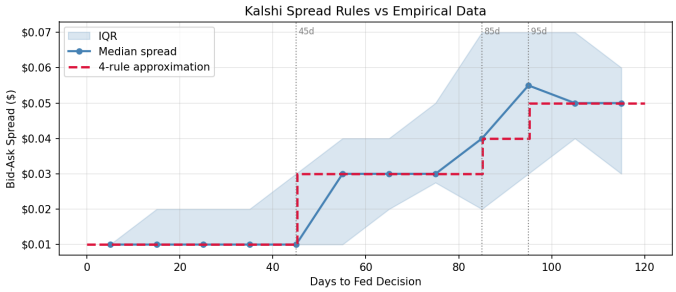


Figure 5: Median Kalshi bid–ask spread as a function of days to decision. The piecewise schedule used in the transaction-cost model is based on this empirical relationship.

- **Transaction fees:** Polymarket currently does not charge an explicit trading fee (free to trade). Kalshi charges a fee of $0.07 \times N \times p \times (1-p)$ per contract, where p is the mid-price and N is the number of contracts (rounded up to whole cents).
- **Total Transaction Cost (one-way):** Transaction Cost (TC) is calculated using: TC = number of contracts · spread + Transaction Fees.

For CME futures legs:

- Market impact cost is modeled as $c = 2 + 15\sqrt{\rho}$ basis points (round-trip), where ρ is the participation rate (traded volume as a fraction of estimated average daily volume for that contract). This formula captures the square-root market-impact law at high participation and a fixed base cost at low participation [9].
- ADV is sourced from the EFFR and SR1 futures volume fields; for dates or contracts without volume records, a conservative default of 100,000 contracts per day is used.

In CME futures, all costs are charged on both entry and exit events. The pipeline emits reproducible artifacts at each stage: merged expectation panels, residual diagnostics (per-asset R^2, half-life, and volatility), OU fit diagnostics, signal trade logs, daily and cumulative P&L, and per-asset and global summary metrics. All random seeds are fixed and no look-ahead is permitted in any estimation or imputation step. We note that relative to prediction markets transaction costs, the CME transaction costs are 10x smaller (TC ≈ 0.01 USD/contract).

Kalshi Market Depth and Position-Size Caps. Volume data is available in the Kalshi data for 39 FOMC meetings spanning May 2023–January 2026. Dollar volume is computed as contracts × actual midprice. Activity is highly concentrated: a small number of days around each FOMC announcement drive the bulk of cumulative volume, while the median active-trading day is far thinner. Kalshi market liquidity has grown substantially through 2025; the three most liquid meetings (Sep, Jul, and May 2025) each recorded total dollar volume exceeding $12M. Table 3 summarises the median daily volume (ADV) by ten-day horizon bin for those three meetings.

Table 3: Kalshi empirical average daily volume by days-to-decision zone (three most liquid meetings: Sep, Jul, May 2025).

DTD Zone	Median ADV (contracts)	Median ADV ($)
40–50 d	∼166,951	∼$48,659
30–40 d	∼307,811	∼$92,070
20–30 d	∼483,600	∼$176,193
10–20 d	∼661,269	∼$218,849
0–10 d	∼2,326,536	∼$596,134

Position-Size Cap Rule. To avoid adverse market impact on the prediction market leg, we impose the following rule of thumb shown in Table 4. These caps apply to the three most liquid meetings; for a typical meeting, pooled-median dollar ADV falls in the range $6,000–$16,000 per day across the 0–40 d horizon, implying proportionally smaller caps.

The strategy's risk-adjusted return metrics (Sharpe ratio, cumulative PnL) are therefore most meaningful at position sizes of \$5,000–\$20,000 per trade for the most liquid meetings, and the reported results should be interpreted as demonstrating *statistical signal quality* rather than scalable capacity.

Table 4: Rule-of-thumb Kalshi position caps (10% of median ADV, 10-day hold; three most liquid meetings).

DTD Zone	**Max Notional**
0–10 d	∼\$59,600
10–20 d	∼\$21,900
20–30 d	∼\$17,600
30–40 d	∼\$9,200
40–50 d	∼\$4,900

8 Conclusion

This report develops a cross-market framework for trading disagreement in FOMC policy expectations between prediction markets and institutional rates instruments. We make two main contributions: a pure arbitrage strategy that constructs matched replicating portfolios on each side and locks in the spread at entry, and a statistical arbitrage strategy that extracts idiosyncratic cross-venue disagreement via PCA and trades mean reversion using Ornstein-Uhlenbeck rules.

The pure arbitrage strategy demonstrates that genuine, repeatable mispricings exist between prediction markets and CME futures. This is consistent with the findings of Diercks, Katz and Wright [6], who establish that Kalshi-implied distributions carry real predictive power for FOMC outcomes, making deviations from futures pricing economically meaningful rather than noise. However, the strategy is not purely mechanical in practice. Whether the spread converges to exactly zero at expiry depends on whether the EFFR tracks the target rate floor consistently, a condition that holds most of the time but not always. In this sense the strategy retains a statistical element: it is a high-confidence convergence bet rather than a locked arbitrage. Transaction costs further limit the opportunity, consuming a substantial fraction of revenue even at minimum position sizes. We also abstract away from liquidity, so the reported win rate and P&L figures should be interpreted as upper bounds on what is achievable in practice rather than definitive performance estimates.

The statistical arbitrage strategy reveals a clear and important finding: a robust gross signal exists in the PCA residual space, confirming that prediction markets and rates markets do not always price the same information in the same way. The PCA decomposition is itself informative: the first component captures the shared macro expectation across all venues, while higher components reflect structural differences in how retail prediction market participants and institutional rates traders form and update their beliefs. However, transaction costs in prediction markets are high enough to fully absorb this signal at any tradable size, rendering the strategy unprofitable net of costs under current market microstructure. This acts to further motivate the use of a pure arbitrage strategy in this framework, where transaction costs are explicitly built into the trading design.

Together these results point to a broader conclusion. Prediction markets are now sufficiently liquid and well-calibrated to be treated as genuine financial instruments. As prediction market liquidity continues to grow, the strategies developed here are likely to become increasingly viable. More broadly, the framework developed in this report provides a reusable tool for decomposing disagreement between expectation markets, with natural extensions to CPI releases, payroll announcements, and other macro events where prediction markets now offer real-money contracts.

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Appendix A Plots

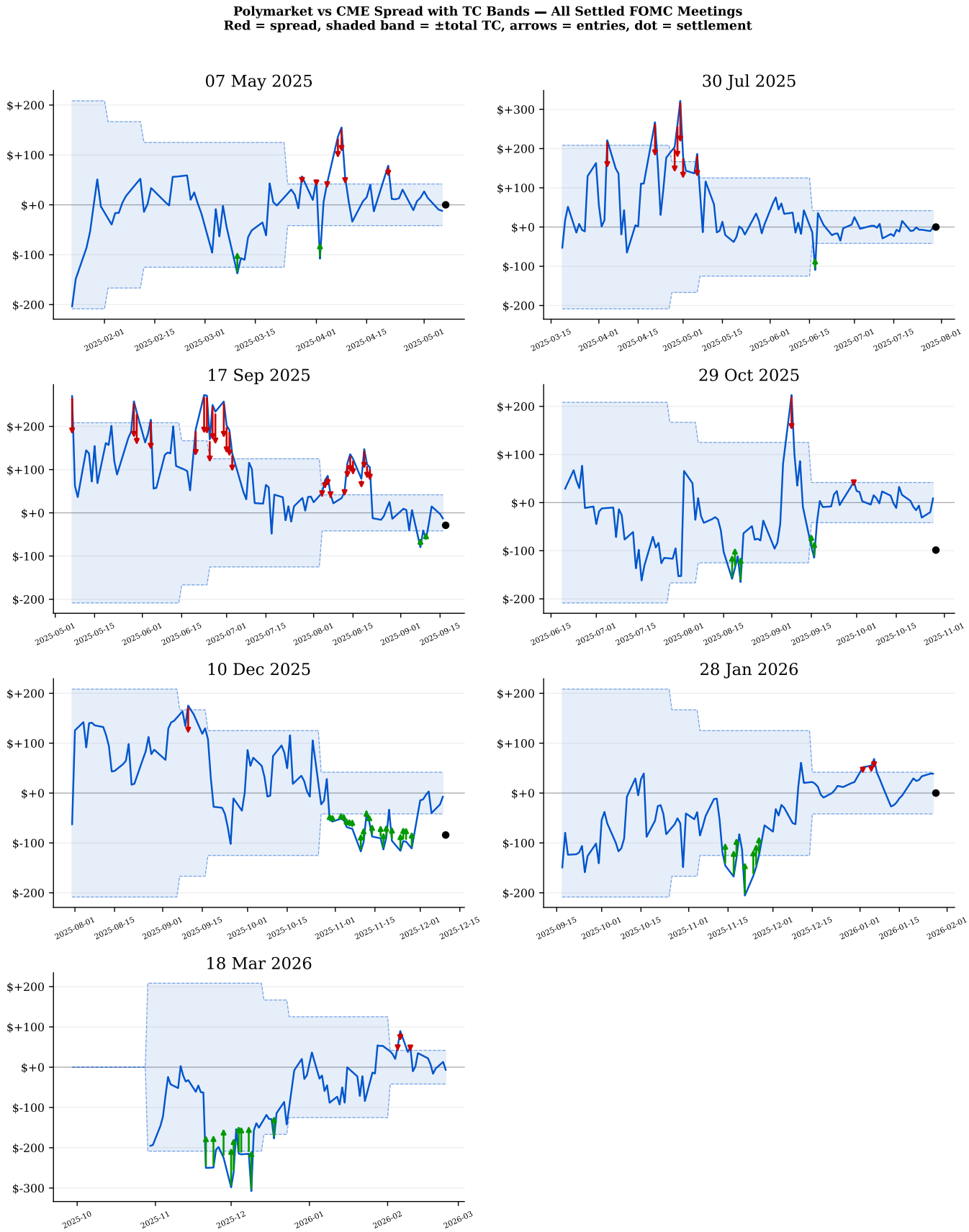

Figure 6: Plot of multiple decision months, for which we had sufficient data for (and of high enough quality). Notice that the March 2026 month is still going on at the time of writing this paper. We observe an expected trend towards zero, although for some months the spread expires off axis.

Appendix B Data

B.1 Instrument Conventions and Assumptions

Prediction-market contracts. Polymarket and Kalshi list binary event contracts on discrete FOMC outcomes. Each contract settles at \$1 if the outcome occurs and \$0 otherwise. Prices are interpreted as risk-neutral probability proxies in $[0, 1]$.

Canonical bucket taxonomy. We map venue-specific labels to a common set of basis-point outcomes: `C50+`, `C25`, `H0`, `H25`, `H50`, `H50+`. This permits direct expected-bps comparison across venues and against rates instruments.

Tail-bucket treatment. Open-ended buckets (`C50+`, `H50+`) are assigned boundary values for expected-value calculations (e.g., 50 bps). This is an explicit approximation: it is accurate when probability mass is concentrated near the boundary and less accurate in rare, large-jump regimes.

Rates instruments and settlement conventions. Fed Funds (`ZQ`) and one-month SOFR (`SR1`) futures settle to monthly averages of overnight rates, not a spot print. Therefore, implied meeting-jump extraction depends on meeting placement within the month (early-month vs late-month meetings). OIS rates are tenor averages that require calendar mapping and business-day alignment before extracting meeting-specific jumps.

B.2 Data, Time Alignment, and Coverage

Primary data inputs

1. **Polymarket FOMC contracts:** timestamped outcome-level prices (raw: 9,071 rows).
2. **Kalshi FOMC contracts:** timestamped candles with bid/ask close fields (raw: 15,759 rows).
3. **CME EFFR futures:** settlements, volume, and open interest for `ZQ`.
4. **CME SOFR futures:** settlements, volume, and open interest for `SR1`.
5. **SOFR OIS tenors:** cross-sectional quotes (ON, 1W, 2W, 1M, 2M, 3M, 6M, 12M).

Appendix C Pure Arbitrage

C.1 Prediction Market Portfolio Example

The effective scaling is DV01 $\times\ \alpha = 41.67 \times 0.5 =$ \$20.83 per basis point:

Outcome h_i (bp)	Contracts	Direction	Cost per contract (\$)	Total (\$)
+50	1,042	Buy	$p_{+50} = 0.05$	52
+25	521	Buy	$p_{+25} = 0.10$	52
0	0	—	—	0
−25	521	Sell	$1 - p_{-25} = 0.30$	156
−50	1,042	Sell	$1 - p_{-50} = 0.85$	886
			Total capital outlay	\$1,146

C.2 Explaining the $(1 - \lambda)$ factor

The main constraint is that we cannot trade fractional futures on CME. A single future (or rather L/S pair) on CME has

$$H = \frac{R_1 - R_0}{1 - \lambda}$$
$$(1 - \lambda)H = R_1 - R_0$$

$(1 - \lambda)H$ is actually the smallest denomination attainable. We can replicate this unit in PM. Then we can scale this trade by any integer multiple.

In practice, market microstructure is similar across decisions and so it is reasonable that we would size our bets roughly equally. If $1 - \lambda < 0.5$, for instance, we could always scale it by an integer c so that $c(1 - \lambda) > 0.5$. Therefore we can assume that our positions are all scaled by a factor of $\gamma \in [0.5, 1]$, so they are all around the same magnitude. For the sake of performance analysis, we neglect this technicality and treat all decisions as if they were sized exactly the same. This makes cumulative PnL a sum of individual decision PnL, which is more convenient.

If we did not take this into account, we would over-state dollar exposure by $1/(1-\lambda)$. The ZQ contract would actually earn $(1-\lambda)\times$ per bp of rate move (not), so all reported dollar P&L figures would be scaled by the partial-month factor:

$$\pi = (1-\lambda) \times \tilde{\pi}, \tag{15}$$

applied to gross P&L, net P&L, TC, and both legs individually. The scaling factor $(1-\lambda) = (D-d)/D$ is fixed per meeting and is listed in Table 5.

Table 5: $(1-\lambda)$ scaling factor per FOMC meeting.

Meeting	Day d	Days D	$1-\lambda$
07 May 2025	7	31	0.774
30 Jul 2025	30	31	0.032
17 Sep 2025	17	30	0.433
29 Oct 2025	29	31	0.065
10 Dec 2025	10	31	0.677
28 Jan 2026	28	31	0.097
18 Mar 2026	18	31	0.419

The signal check and portfolio construction are performed in full-H space (unscaled) so that the trading logic is independent of the accident of which day of the month the FOMC meets.

Net P&L with correction (not used in our backtesting):

$$\pi^{\text{net}} = (1-\lambda)\left(\tilde{\pi}^{\text{PM}} + \tilde{\pi}^{\text{CME}} - \text{TC}\right). \tag{16}$$

Appendix D Statistical Arbitrage

D.1 SR1 and OIS Estimation

Regularized estimation

Parameters are estimated via

$$\min_{\theta_t} \|A_t\theta_t - y_t\|_2^2 + \lambda\|\Gamma\theta_t\|_2^2,$$

with $\lambda = 10^{-6}$ and Γ penalizing jump parameters but not baseline r_0. This is purely numerical stabilization under near-collinear exposures.

SR1 implementation

For SR1, each row corresponds to a calendar-month window $[s_i, e_i]$:

$$A_{i,k} = \frac{\#\text{business days in } [\max(s_i, m_k+1), e_i]}{\#\text{business days in } [s_i, e_i]},$$

and instrument rate is $y_i = (100 - \text{settlement}_i)/100$. Outputs are stored as expected rate changes. Meeting-jump portfolio weights are extracted from the relevant pseudoinverse rows and stored accordingly.

OIS implementation

For OIS, tenor codes are parsed into unit/value, maturity dates are computed with T+2 spot lag and modified-following calendar rules, and ACT/360 fractions define tenor lengths. OIS tenors are capped at 6 months for local meeting-jump identification. Outputs are expected jump and portfolio weights following SR1.

D.2 PCA Details

Causal imputation and residualization

Missing values are imputed on a strictly backward-looking basis: forward carry within meeting, then historical-mean backfill using only prior observations. PCA is fit on centered features with $K = 3$ components, and trading residuals remove the first one:

$$\hat{u}_{j,t} = x_{j,t} - \hat{\mu}_j - \hat{\lambda}_{j1} f_{1,t}.$$

Rolling fit quality:

$$R_t^2 = 1 - \frac{\overline{u_{j,t}^2}}{\overline{x_{j,t}^2}},$$

with finite values reported only after at least 5 observations.

D.3 Stat-Arb OU Rules and Filters

OU estimation

Residuals are modeled with rolling AR(1):

$$u_{t+1} = a + bu_t + \epsilon_{t+1}, \quad \kappa = -\ln b, \quad \mu = \frac{a}{1-b}, \quad \sigma_{OU} = \sigma_\epsilon \sqrt{\frac{2\kappa}{1-b^2}}.$$

Estimates are shifted by one period before trading. Invalid estimates are discarded when $b \leq 0$, $b \geq 1 - \varepsilon$ ($\varepsilon = 10^{-8}$), or $\sigma_{OU} \leq 0$.

Entry/exit and risk controls

- Entry long: $u_t \leq \mu - 1.25\sigma_{OU}$.
  - Sensitivity: $u_t \leq \mu - 2.5\sigma_{OU}$.
- Entry short: $u_t \geq \mu + 1.25\sigma_{OU}$.
  - Sensitivity: $u_t \geq \mu + 2.5\sigma_{OU}$.
- Exit long: $u_t \geq \mu - 0.5\sigma_{OU}$.
- Exit short: $u_t \leq \mu + 0.5\sigma_{OU}$.
- Forced flat: close all positions when $t \geq m - 2$ days.

D.4 Sensitivities

To assess strategy sensitivity to data granularity, we evaluate four feature representations (modes). Each mode integrates institutional rates (EFFR, SR1, OIS) with varying transformations of Polymarket and Kalshi probability distributions:

Expected: The most aggregated mode; collapses the distribution into a single expected basis-point move per venue. It prioritizes stability but obscures distributional shape and market uncertainty.

Grouped: Partitions the expected move into total "hike" and "cut" basis points. This default specification balances directional granularity with robustness against noise in thinly traded tail buckets.

Moments: Includes the expected move and the variance across outcome buckets. By parameterizing dispersion, it allows the PCA to isolate market uncertainty as a factor distinct from directional expectations.

All: The most granular mode; treats each individual outcome bucket (e.g., H25, C25) as a unique probability-weighted product. While capturing the full discrete distribution, it is highly susceptible to noise and collinearity from sparse order books.

D.5 Stat-Arb Execution and P&L Details

Two-layer mapping

Signal positions in residual units are converted to underlyings by:

1. PCA residual weights:
$$w^{(j)} = e_j - \hat{\Lambda}_1 \hat{\Lambda}_1' e_j.$$
2. Feature-to-underlying mapping: EFFR contract mapping, SR1/OIS portfolio weights from matrix estimators, and prediction-bucket mappings via canonical bps map.

Combined holding for instrument ℓ:

$$q_{j,\ell} = \sum_{n=1}^{N} w_n^{(j)} m_{n,\ell}.$$

Execution timing and mark-to-market

Default execution lag is 0 (same-day close), with lag 1 used in sensitivity analysis for robustness. Signals without valid execution dates before blackout are dropped. Daily P&L for active positions:

$$\Pi_{j,t} = \sum_{\ell} q_{j,\ell}(P_{\ell,t} - P_{\ell,\tau}),$$

aggregated across positions/meetings into portfolio daily series, cumulative P&L, drawdown, and Sharpe.